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SAAT / GAT QUANTITATIVE (ENGLISH) • UNIT 7.01

SAAT Trigonometry, Conic Sections & 3D Vectors

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 D	Q3 A	Q4 B	Q5 C
Q6 D	Q7 A	Q8 B	Q9 D	Q10 D
Q11 B	Q12 B	Q13 A	Q14 A	Q15 D
Q16 D	Q17 B	Q18 B	Q19 A	Q20 B
Q21 C	Q22 A	Q23 D	Q24 D	Q25 D

SAAT Trigonometry, Conic Sections & 3D Vectors Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 18x + 98$ in the domain of SAAT Trigonometry, Conics & Vectors. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(11)$.



A 47

B 21

C 38 ✓

D 36

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Trigonometry, Conics & Vectors):

1. Complete the square: $x^2 - 18x + 98 = (x-9)^2 - 81 + 98 = (x-9)^2 + 17$, giving minimum value $k = 17$. **[M1]**
2. Evaluate at $x = 11$: $f_1(11) = (2)^2 + 17 = 21$. **[M1]**
3. Sum the two quantities: $k + f_1(11) = 17 + 21 = 38$. **[A1]**

- **Final Verified Answer:** 38.

2 MCQ

A mathematical model in SAAT Trigonometry, Conics & Vectors is defined by the cubic function $y_2(x) = 2x^3 - 9x^2 + 15x - 14$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 69

B 0

C $15 + 1$

D 15 ✓

WORKED SOLUTION & VERIFICATION

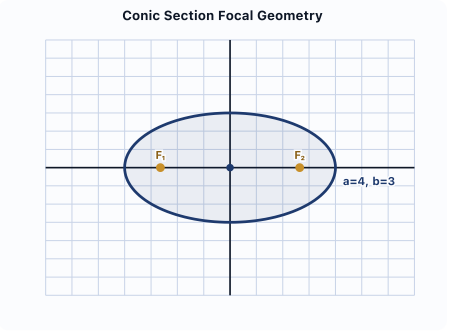
Mark Scheme & Examiner Solution (SAAT Trigonometry, Conics & Vectors):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 6x^2 - 18x + 15$. **[M1]**
2. Substitute $x = 3$: $\left.\frac{dy}{dx}\right|_{x=3} = 6(9) - 18(3) + 15 = 15$. **[M1][A1]**

- **Final Verified Answer:** 15.

3 MCQ

In SAAT Trigonometry, Conics & Vectors, two displacement vectors are given by $\mathbf{u}_3 = \langle 9, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.



A 170 ✓

B 25

C 145

D 282

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Trigonometry, Conics & Vectors):

1. Compute the inner product: $\mathbf{u}_3 \cdot \mathbf{v}_3 = (9)(9) + (-8)(7) = 81 + (-56) = 25$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_3\|^2 = 9^2 + (-8)^2 = 145$. **[M1]**
3. Combine: $25 + 145 = 170$. **[A1]**

- **Final Verified Answer:** 170.

4 MCQ

The linear growth profile shown in the diagram for SAAT Trigonometry, Conics & Vectors follows an arithmetic progression with first term $a_1 = 10$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 94

B 780 ✓

C 870

D 1410

WORKED SOLUTION & VERIFICATION

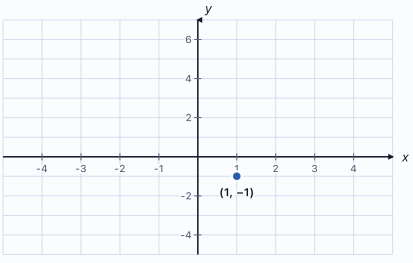
Mark Scheme & Examiner Solution (SAAT Trigonometry, Conics & Vectors):

1. Use the sum formula for an arithmetic series: $S_n = \frac{n}{2}[2a_1 + (n-1)d]$. **[M1]**
2. Substitute $n = 15, a_1 = 10, d = 6$: $S_{15} = \frac{15}{2}[20 + 14 \times 6] = \frac{15}{2} \times 104 = 780$. **[M1][A1]**

- **Final Verified Answer:** 780.

5 MCQ

In SAAT Trigonometry, Conics & Vectors, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 36x)\,\mathrm{d}x$.



A 144

B 156

C 120 ✓

D 61

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Trigonometry, Conics & Vectors):

1. Integrate term by term: $\int (18x^2 + 36x)\,\mathrm{d}x = [6x^3 + 18x^2]_0^2$. **[M1]**
2. Substitute the upper and lower limits: $(6(2)^3 + 18(2)^2) - 0 = 48 + 72 = 120$. **[M1][A1]**

- **Final Verified Answer:** 120.

6 MCQ

In SAAT Trigonometry, Conics & Vectors, consider the quadratic function $f_6(x) = x^2 - 20x + 189$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(17)$.

A 237

B 138

C 220

D 227 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Trigonometry, Conics & Vectors):

1. Complete the square: $x^2 - 20x + 189 = (x-10)^2 - 100 + 189 = (x-10)^2 + 89$, giving minimum value $k = 89$. **[M1]**

2. Evaluate at $x = 17$: $f_6(17) = (7)^2 + 89 = 138$. **[M1]**

3. Sum the two quantities: $k + f_6(17) = 89 + 138 = 227$. **[A1]**

- **Final Verified Answer:** 227.

7 MCQ

The graph illustrates the cubic model $y_7(x) = 2x^3 - 10x^2 + 33x - 21$ studied in SAAT Trigonometry, Conics & Vectors. Find the instantaneous gradient $\frac{dy}{dx}$ of the tangent to the curve at $x = 8$.

A 257 ✓

B 417

C 224

D 61

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Trigonometry, Conics & Vectors):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 6x^2 - 20x + 33$. **[M1]**

2. Substitute $x = 8$: $\left.\frac{dy}{dx}\right|_{x=8} = 6(64) - 20(8) + 33 = 257$. **[M1][A1]**

- **Final Verified Answer:** 257.

8 MCQ

In SAAT Trigonometry, Conics & Vectors, two displacement vectors are given by $\mathbf{u}_8 = \langle 10, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 45

B 194 ✓

C 149

D 264

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Trigonometry, Conics & Vectors):

1. Compute the inner product: $\mathbf{u}_8 \cdot \mathbf{v}_8 = (10)(8) + (-7)(5) = 80 + (-35) = 45$. **[M1]**

2. Compute the squared norm: $\|\mathbf{u}_8\|^2 = 10^2 + (-7)^2 = 149$. **[M1]**

3. Combine: $45 + 149 = 194$. **[A1]**

- **Final Verified Answer:** 194.

9 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 2, k \rangle$ and $\mathbf{v} = \langle -3, 3 \rangle$ orthogonal (perpendicular)?

A -2	B 5
C 14	D 2 ✓

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(2)(-3) + k(3) = 0 \implies 3k = 6 \implies k = 2$.

10 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 3, k \rangle$ and $\mathbf{v} = \langle -8, 4 \rangle$ orthogonal (perpendicular)?

A -6	B 10
C 7	D 6 ✓

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(3)(-8) + k(4) = 0 \implies 4k = 24 \implies k = 6$.

11 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 4, k \rangle$ and $\mathbf{v} = \langle -15, 5 \rangle$ orthogonal (perpendicular)?

A 17	B 12 ✓
C 9	D -12

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(4)(-15) + k(5) = 0 \implies 5k = 60 \implies k = 12$.

12 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 5, k \rangle$ and $\mathbf{v} = \langle -24, 6 \rangle$ orthogonal (perpendicular)?

A -20	B 20 ✓
C 26	D 11

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(5)(-24) + k(6) = 0 \implies 6k = 120 \implies k = 20$.

13 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 6, k \rangle$ and $\mathbf{v} = \langle -35, 7 \rangle$ orthogonal (perpendicular)?

A 30 ✓	B 37
C -30	D 13

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(6)(-35) + k(7) = 0 \implies 7k = 210 \implies k = 30$.

14 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 7, k \rangle$ and $\mathbf{v} = \langle -2, 2 \rangle$ orthogonal (perpendicular)?

- **A** 7 ✓
- **B** -7
- **C** 14
- **D** 9

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(7)(-2) + k(2) = 0 \implies 2k = 14 \implies k = 7$.

15 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 8, k \rangle$ and $\mathbf{v} = \langle -6, 3 \rangle$ orthogonal (perpendicular)?

- **A** 11
- **B** -16
- **C** 19
- **D** 16 ✓

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(8)(-6) + k(3) = 0 \implies 3k = 48 \implies k = 16$.

16 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 2, k \rangle$ and $\mathbf{v} = \langle -12, 4 \rangle$ orthogonal (perpendicular)?

- **A** 10
- **B** 14
- **C** -6
- **D** 6 ✓

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(2)(-12) + k(4) = 0 \implies 4k = 24 \implies k = 6$.

17 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 3, k \rangle$ and $\mathbf{v} = \langle -20, 5 \rangle$ orthogonal (perpendicular)?

- **A** 8
- **B** 12 ✓
- **C** 17
- **D** -12

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(3)(-20) + k(5) = 0 \implies 5k = 60 \implies k = 12$.

18 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 4, k \rangle$ and $\mathbf{v} = \langle -30, 6 \rangle$ orthogonal (perpendicular)?

- **A** -20
- **B** 20 ✓
- **C** 10
- **D** 26

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(4)(-30) + k(6) = 0 \implies 6k = 120 \implies k = 20$.

19 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 5, k \rangle$ and $\mathbf{v} = \langle -7, 7 \rangle$ orthogonal (perpendicular)?

- **A** 5 ✓
- **B** 14
- **C** -5
- **D** 12

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(5)(-7) + k(7) = 0 \implies 7k = 35 \implies k = 5$.

20 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 6, k \rangle$ and $\mathbf{v} = \langle -4, 2 \rangle$ orthogonal (perpendicular)?

- **A** 8
- **B** 12 ✓
- **C** 14
- **D** -12

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(6)(-4) + k(2) = 0 \implies 2k = 24 \implies k = 12$.

21 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 7, k \rangle$ and $\mathbf{v} = \langle -9, 3 \rangle$ orthogonal (perpendicular)?

- **A** 24
- **B** 10
- **C** 21 ✓
- **D** -21

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(7)(-9) + k(3) = 0 \implies 3k = 63 \implies k = 21$.

22 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 8, k \rangle$ and $\mathbf{v} = \langle -16, 4 \rangle$ orthogonal (perpendicular)?

- **A** 32 ✓
- **B** -32
- **C** 36
- **D** 12

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(8)(-16) + k(4) = 0 \implies 4k = 128 \implies k = 32$.

23 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 2, k \rangle$ and $\mathbf{v} = \langle -25, 5 \rangle$ orthogonal (perpendicular)?

- **A** -10
- **B** 7
- **C** 15
- **D** 10 ✓

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(2)(-25) + k(5) = 0 \implies 5k = 50 \implies k = 10$.

24 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 3, k \rangle$ and $\mathbf{v} = \langle -6, 6 \rangle$ orthogonal (perpendicular)?

A 9

B -3

C 14

D 3 ✓

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(3)(-6) + k(6) = 0 \implies 6k = 18 \implies k = 3$.

25 MCQ

For what value of k are the vectors $\mathbf{u} = \langle 4, k \rangle$ and $\mathbf{v} = \langle -14, 7 \rangle$ orthogonal (perpendicular)?

A -8

B 11

C 15

D 8 ✓

WORKED SOLUTION & VERIFICATION

Orthogonal vectors satisfy $\mathbf{u} \cdot \mathbf{v} = 0$: $(4)(-14) + k(7) = 0 \implies 7k = 56 \implies k = 8$.